Quantum Mechanics II Excercise 3

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1 $SU(2), SO(3)$

1.1 $SU(2)$ and the Pauli matrices

$SU(2)$ is defined as the set of all 2×2 complex matrices such that $U^\dagger U = \mathbb{1}$ and $\det U = 1$. We can characterise those matrices by starting from a generic 2×2 matrix and narrowing down using the conditions. Given a 2×2 matrix U, it can be written as

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{1}$$

The first condition for it to be in $SU(2)$ is $U^\dagger U = \mathbb{1}$. This means that the matrix is invertible and that its inverse is its complex conjugate

$$U^\dagger = U^{-1} \tag{2}$$

Using the formula for the inverse of a 2×2 matrix:

$$U^{-1} = \frac{1}{|\det U|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{3}$$

And using the second condition $\det U = 1$ we get:

$$\begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{4}$$

$$a^* = d \qquad d^* = a \qquad b^* = -c \qquad c^* = -b \tag{5}$$

In other words, we can write U as:

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \tag{6}$$

Writing a and b in their complex cartesian representation, we get:

$$U = \begin{pmatrix} x + iy & w + iz \\ -w + iz & x - iy \end{pmatrix} \tag{7}$$

And requiring $\det U = 1$ gives us:

$$x^2 + y^2 + w^2 + z^2 = 1 \tag{8}$$

Now, let us look at the expression

$$A = \exp\left[it_i\sigma_i\right] \tag{9}$$

Using the definition of the Pauli matrices, we get

$$A = \exp\left[\frac{1}{2}i\theta_i\sigma_i\right] = \exp\left[\frac{1}{2}\begin{pmatrix} i\theta_z & \theta_y + i\theta_x \\ -\theta_y + i\theta_x & -i\theta_z \end{pmatrix}\right] \equiv \exp\left[\frac{1}{2}P(\theta)\right] \tag{10}$$

Where we named the matrix in the exponent $P(\theta)$. We can now expand the exponent, but first let us calculate the powers of $P(\theta)$:

$$P^2(\theta) = \begin{pmatrix} i\theta_z & \theta_y + i\theta_x \\ -\theta_y + i\theta_x & -i\theta_z \end{pmatrix} \begin{pmatrix} i\theta_z & \theta_y + i\theta_x \\ -\theta_y + i\theta_x & -i\theta_z \end{pmatrix} = -\begin{pmatrix} \theta_x^2 + \theta_y^2 + \theta_z^2 & 0 \\ 0 & \theta_x^2 + \theta_y^2 + \theta_z^2 \end{pmatrix} \equiv -\theta^2 \mathbb{1} \quad (11)$$

Therefore, we can split the powers of P to even and odd powers:

$$P^{2n}(\theta) = (-1)^n \, \theta^{2n} \mathbb{1} \qquad P^{2n+1}(\theta) = (-1)^n \, \theta^{2n} P \quad (12)$$

And thus

$$\begin{aligned} A = \exp\left[\frac{1}{2}P\right] &= \sum_{n=0}^{\infty} \frac{(P/2)^n}{n!} = \mathbb{1} \sum_{n \ even} \frac{(i\theta/2)^n}{n!} + \frac{1}{2}P \sum_{n \ odd} \frac{(i\theta/2)^{n-1}}{n!} = \cos\left(\frac{\theta}{2}\right)\mathbb{1} + \sin\left(\frac{\theta}{2}\right)\frac{P}{\theta} = \\ &= \begin{pmatrix} \cos(\theta/2) + i\sin(\theta/2)\frac{\theta_z}{\theta} & \sin(\theta/2)\frac{\theta_y + i\theta_x}{\theta} \\ \sin(\theta/2)\frac{-\theta_y + i\theta_x}{\theta} & \cos(\theta/2) - i\sin(\theta/2)\frac{\theta_z}{\theta} \end{pmatrix} \end{aligned} \quad (13)$$

The determinant of this matrix is:

$$\begin{aligned} \det A &= \left(\cos(\theta/2) + i\sin(\theta/2)\frac{\theta_z}{\theta}\right)\left(\cos(\theta/2) - i\sin(\theta/2)\frac{\theta_z}{\theta}\right) - \sin^2(\theta/2)\frac{-\theta_x^2 - \theta_y^2}{\theta^2} = \\ &= \cos^2(\theta/2) + \sin^2(\theta/2)\frac{\theta_z^2}{\theta^2} + \sin^2(\theta/2)\frac{\theta_x^2 + \theta_y^2}{\theta^2} = \cos^2(\theta/2) + \sin^2(\theta/2)\frac{\theta_x^2 + \theta_y^2 + \theta_z^2}{\theta^2} = \\ &= \cos^2(\theta/2) + \sin^2(\theta/2) = 1 \end{aligned} \quad (14)$$

Which means that with a small change of variables we can make it fit the matrix in equation 7:

$$x = \cos(\theta/2) \qquad y = \sin(\theta/2)\frac{\theta_z}{\theta} \qquad w = \sin(\theta/2)\frac{\theta_y}{\theta} \qquad z = \sin(\theta/2)\frac{\theta_x}{\theta} \quad (15)$$

Which means that for every 2×2 matrix U in $SU(2)$ one can find $\theta_x, \theta_y, \theta_z$ such that $U = \exp\left[\frac{1}{2}i\theta_i\sigma_i\right]$, and thus $t_i = \frac{1}{2}\sigma_i$ furnish a representation of the $su(2)$ algebra.

1.2 The $SO(3)$ group

The three matrices that make up the representation of the algebra are:

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \qquad L_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \qquad L_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (16)$$

Their linear sum with constant $-i\varphi_i$ is

$$P = (-i)i \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} \quad (17)$$

Which means that

$$A = \exp\left[-i\varphi_i L_i\right] = \exp P = \exp \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} \quad (18)$$

To show that every matrix in $SO(3)$ has the form of A, we need to open the exponent. The powers of P are:

$$P^2 = \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} = \begin{pmatrix} -\varphi_2^2 - \varphi_3^2 & \varphi_1\varphi_2 & \varphi_1\varphi_3 \\ \varphi_1\varphi_2 & -\varphi_1^2 - \varphi_3^2 & \varphi_2\varphi_3 \\ \varphi_1\varphi_3 & \varphi_2\varphi_3 & -\varphi_1^2 - \varphi_2^2 \end{pmatrix} \quad (19)$$

$$P^3 = \begin{pmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{pmatrix} \begin{pmatrix} -\varphi_2^2 - \varphi_3^2 & \varphi_1\varphi_2 & \varphi_1\varphi_3 \\ \varphi_1\varphi_2 & -\varphi_1^2 - \varphi_3^2 & \varphi_2\varphi_3 \\ \varphi_1\varphi_3 & \varphi_2\varphi_3 & -\varphi_1^2 - \varphi_2^2 \end{pmatrix} = -\varphi^2 P \tag{20}$$

Then, for $n > 0$

$$P^{2n} = (i\varphi)^{2n-2} P^2 \qquad P^{2n+1} = (i\varphi)^{2n} P \tag{21}$$

$$\begin{aligned} A = \exp P = \sum_{n=0}^{\infty} \frac{P^n}{n!} = \sum_{n\ even} \frac{P^n}{n!} + \sum_{n\ odd} \frac{P^n}{n!} = \mathbb{1} + \sum_{n>0\ even} \frac{(i\varphi)^{n-1}}{n!} P^2 + \sum_{n\ odd} \frac{(i\varphi)^{n-1}}{n!} P = \\ = \mathbb{1} + \sin(\varphi)\frac{P}{\varphi} + (1-\cos\varphi)\,\frac{P^2}{\varphi^2} = \\ = \begin{pmatrix} \cos\varphi + n_1^2\,(1-\cos\varphi) & n_1n_2\,(1-\cos\varphi) - n_3\sin\varphi & n_1n_3\,(1-\cos\varphi) + n_2\sin\varphi \\ n_1n_2\,(1-\cos\varphi) - n_3\sin(\varphi) & \cos(\varphi) + n_2^2\,(1-\cos(\varphi)) & n_2n_3\,(1-\cos\varphi) - n_1\sin(\varphi) \\ n_1n_3\,(1-\cos\varphi) - n_2\sin(\varphi) & n_2n_3\,(1-\cos\varphi) + n_1\sin(\varphi) & \cos(\varphi) + n_3^2\,(1-\cos(\varphi)) \end{pmatrix} \end{aligned} \tag{22}$$

Where $n_i = \varphi_i/\varphi$. This is Rodrigues' rotation formula, which is the most general form of a 3D rotation matrix.
Now, for any 3×3 matrix O we can write:

$$O = \begin{pmatrix} a & b & c \\ d & e & f \\ g & j & h \end{pmatrix} \tag{23}$$

And if it's in $SO(3)$, $O^T = O^{-1}$, we can use the generic formula $A^{-1} = \frac{1}{|\det A|}Adj(A)$ and the fact that $\det O = 1$ to get

$$O^T = O^{-1} = \frac{1}{|\det O|}Adj(O) = Adj(O) \tag{24}$$

This gives us the formulas:

$$\begin{pmatrix} a \\ d \\ g \end{pmatrix} = \begin{pmatrix} eh - fj \\ cj - bh \\ bf - ce \end{pmatrix} = \begin{pmatrix} c \\ f \\ h \end{pmatrix} \times \begin{pmatrix} b \\ e \\ j \end{pmatrix} \tag{25}$$

$$\begin{pmatrix} b \\ e \\ j \end{pmatrix} = \begin{pmatrix} a \\ d \\ g \end{pmatrix} \times \begin{pmatrix} c \\ f \\ h \end{pmatrix} \qquad \begin{pmatrix} b \\ e \\ j \end{pmatrix} = \begin{pmatrix} a \\ d \\ g \end{pmatrix} \times \begin{pmatrix} c \\ f \\ h \end{pmatrix} \tag{26}$$

In other words, the three columns of the matrix are an orthonormal basis of $\mathbb{R}^3$ (what a surprise! It was definity not in the group's name or anything). Every orthonormal basis can be displayed as a rotation from some preselected reference base. This means the $SO(3)$ matrices are rotation matrices, and can therefore be written in the form of the Rodrigues formula.

1.3 Commutation relations

It's easy to see that for $t_i = \frac{1}{2}\sigma_i$, $[t_i, t_j] = i\varepsilon_{ijk}t_k$. The simplest way is just to check them one by one:

$$
\begin{gathered}
[t_1,t_1]=0 \qquad [t_2,t_2]=0 \qquad [t_3,t_3]=0\\
[t_1,t_2]=-[t_2,t_1]=t_1t_2-t_2t_1=\\
=\frac{1}{4}\left[\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}0&-i\\i&0\end{pmatrix}-\begin{pmatrix}0&-i\\i&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}\right]=\frac{1}{4}\left[\begin{pmatrix}i&0\\0&-i\end{pmatrix}-\begin{pmatrix}-i&0\\0&i\end{pmatrix}\right]=\\
=\frac{1}{4}\left[i\sigma_3-(-i)\,\sigma_3\right]=\frac{1}{2}i\sigma_3=it_3\\
[t_1,t_3]=-[t_3,t_1]=t_1t_3-t_3t_1=\\
=\frac{1}{4}\left[\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix}-\begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}\right]=\frac{1}{4}\left[\begin{pmatrix}0&-1\\1&0\end{pmatrix}-\begin{pmatrix}0&1\\-1&0\end{pmatrix}\right]=\\
=\frac{1}{4}\left[-i\sigma_2+(-i)\,\sigma_2\right]=-\frac{1}{2}i\sigma_2=-it_2\\
[t_2,t_3]=-[t_3,t_2]=t_2t_3-t_3t_2=\\
=\frac{1}{4}\left[\begin{pmatrix}0&-i\\i&0\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix}-\begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}0&-i\\i&0\end{pmatrix}\right]=\frac{1}{4}\left[\begin{pmatrix}0&i\\i&0\end{pmatrix}-\begin{pmatrix}0&-i\\-i&0\end{pmatrix}\right]=\\
=\frac{1}{4}\left[i\sigma_1-(-i)\,\sigma_1\right]=\frac{1}{2}i\sigma_1=it_1
\end{gathered}
\tag{27}
$$

Similarly, for L_i we get

$$
\begin{gathered}
[L_1,L_1]=0 \qquad [L_2,L_2]=0 \qquad [L_3,L_3]=0\\
[L_1,L_2]=-[L_2,L_1]=L_1L_2-L_2L_1=\\
=\begin{pmatrix}0&0&0\\0&0&-i\\0&i&0\end{pmatrix}\begin{pmatrix}0&0&i\\0&0&0\\-i&0&0\end{pmatrix}-\begin{pmatrix}0&0&i\\0&0&0\\-i&0&0\end{pmatrix}\begin{pmatrix}0&0&0\\0&0&-i\\0&i&0\end{pmatrix}=\\
=\begin{pmatrix}0&0&0\\-1&0&0\\0&0&0\end{pmatrix}-\begin{pmatrix}0&-1&0\\0&0&0\\0&0&0\end{pmatrix}=i\begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix}=iL_3\\
[L_1,L_3]=-[L_3,L_1]=L_1L_3-L_3L_1=\\
=\begin{pmatrix}0&0&0\\0&0&-i\\0&i&0\end{pmatrix}\begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix}-\begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix}\begin{pmatrix}0&0&0\\0&0&-i\\0&i&0\end{pmatrix}=\\
=\begin{pmatrix}0&0&0\\0&0&0\\-1&0&0\end{pmatrix}-\begin{pmatrix}0&0&-1\\0&0&0\\0&0&0\end{pmatrix}=i\begin{pmatrix}0&0&-i\\0&0&0\\i&0&0\end{pmatrix}=-iL_2\\
[L_2,L_3]=-[L_3,L_2]=L_2L_3-L_3L_2=\\
=\begin{pmatrix}0&0&i\\0&0&0\\-i&0&0\end{pmatrix}\begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix}-\begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix}\begin{pmatrix}0&0&i\\0&0&0\\-i&0&0\end{pmatrix}=\\
=\begin{pmatrix}0&0&0\\0&0&0\\0&-1&0\end{pmatrix}-\begin{pmatrix}0&0&0\\0&0&-1\\0&0&0\end{pmatrix}=i\begin{pmatrix}0&0&0\\0&0&-i\\0&i&0\end{pmatrix}=iL_1
\end{gathered}
\tag{28}
$$

This means that both t_i and L_i satisfy the same commutation relations, and there is an obvious isomorphism between their respective algebras.

1.4 Equivilance of transformations in $SU(2)$ and $SO(3)$

For every $x_i \in \mathbb{R}^3$, let the 2×2 matrix be

$$X = x_i \cdot \sigma_i = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_3 \end{pmatrix} \tag{29}$$

Now, using equation 13, we can transform X as such:

$$U = \cos(\theta/2)\mathbb{1} + \sin(\theta/2)\frac{P}{\theta} \qquad P = \begin{pmatrix} i\theta_z & \theta_y + i\theta_x \\ -\theta_y + i\theta_x & -i\theta_z \end{pmatrix} = i\sum_k \theta_k \sigma_k \tag{30}$$

$$\begin{gathered} U^\dagger X U = \\ = \left[\cos(\theta/2)\mathbb{1} + \sin(\theta/2)\frac{P}{\theta}\right]^\dagger X \left[\cos(\theta/2)\mathbb{1} + \sin(\theta/2)\frac{P}{\theta}\right] = \\ = \cos^2(\theta/2)X + \theta^{-1}\sin(\theta/2)\cos(\theta/2)[P^\dagger X - XP] + \theta^{-2}\sin^2(\theta/2)P^\dagger XP \end{gathered} \tag{31}$$

Now we can open each term using $\sigma_i\sigma_j = i\varepsilon_{ijk}\sigma_k + \delta_{ij}\mathbb{1}$ and $\sigma^\dagger = \sigma$:

$$\begin{gathered} XP = (x_i\sigma_i)(i\theta_j\sigma_j) = ix_i\theta_j\sigma_i\sigma_j = ix_i\theta_j(i\varepsilon_{ijk}\sigma_k + \delta_{ij}\mathbb{1}) = -x_i\theta_j\varepsilon_{ijk}\sigma_k + ix_i\theta_i\mathbb{1} \\ P^\dagger X = (i\theta_j\sigma_j^\dagger)(x_i\sigma_i) = ix_i\theta_j\sigma_j\sigma_i = ix_i\theta_j(-i\varepsilon_{ijk}\sigma_k + \delta_{ij}\mathbb{1}) = x_i\theta_j\varepsilon_{ijk}\sigma_k + ix_i\theta_i\mathbb{1} \\ P^\dagger X - XP = -2x_i\theta_j\varepsilon_{ijk}\sigma_k = 2(\vec{\theta} \times \vec{x}) \cdot \vec{\sigma} \\ P^\dagger XP = (x_i\theta_j\varepsilon_{ijk}\sigma_k + ix_i\theta_i\mathbb{1})(i\theta_l\sigma_l) = ix_i\theta_j\theta_l\varepsilon_{ijk}\sigma_k\sigma_l - x_i\theta_i\theta_l\sigma_l = \\ = ix_i\theta_j\theta_l\varepsilon_{ijk}(i\varepsilon_{klm}\sigma_m + \delta_{kl}\mathbb{1}) - x_i\theta_i\theta_l\sigma_l = \\ = -x_i\theta_j\theta_l\varepsilon_{ijk}\varepsilon_{klm}\sigma_m + ix_i\theta_j\theta_l\varepsilon_{ijk}\delta_{kl}\mathbb{1} - x_i\theta_i\theta_l\sigma_l = \\ = -x_i\theta_j\theta_l(\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\sigma_m + ix_i\theta_j\theta_l\varepsilon_{ijl}\mathbb{1} - x_i\theta_i\theta_l\sigma_l \end{gathered} \tag{32}$$

The third term this equaiton is a product of a symmetric tensor $x_i\theta_j\theta_l$ and an antisymmetric tensor ε_{ijl}, which means it's zero. This leaves:

$$P^\dagger XP = -x_i\theta_j\theta_i\sigma_j + x_i\theta_j\theta_j\sigma_i - x_i\theta_i\theta_l\sigma_l = \theta^2\vec{x}\cdot\vec{\sigma} - 2(\vec{x}\cdot\vec{\theta})(\vec{\theta}\cdot\vec{\sigma}) \tag{33}$$

In total, we get

$$\begin{gathered} U^\dagger XU = \\ = \cos^2(\theta/2)X + \theta^{-1}\sin(\theta/2)\cos(\theta/2)\cdot 2(\vec{\theta}\times\vec{x})\cdot\vec{\sigma} + \theta^{-2}\sin^2(\theta/2)\left(\theta^2\vec{x}\cdot\vec{\sigma} - 2(\vec{x}\cdot\vec{\theta})(\vec{\theta}\cdot\vec{\sigma})\right) = \\ = \cos^2(\theta/2)X + 2\theta^{-1}\sin(\theta/2)\cos(\theta/2)(\vec{\theta}\times\vec{x})\cdot\vec{\sigma} + \sin^2(\theta/2)X - 2\theta^{-2}\sin^2(\theta/2)(\vec{x}\cdot\vec{\theta})(\vec{\theta}\cdot\vec{\sigma}) = \\ = X + \theta^{-1}\sin(\theta)(\vec{\theta}\times\vec{x})\cdot\vec{\sigma} - \theta^{-2}(1-\cos(\theta))(\vec{\theta}\cdot\vec{x})(\vec{\theta}\cdot\vec{\sigma}) = \\ = \left(x_i + \theta^{-1}\sin(\theta)(\varepsilon_{jki}\theta_j x_k) - \theta^{-2}(1-\cos(\theta))\theta_j\theta_i x_j\right)\sigma_i \end{gathered} \tag{34}$$

defining $L_{ij} = \varepsilon_{kij}\theta_k$, we get:

$$L^2 x = L \times (L \times x) = L(Lx) = \theta_i\theta_j x_j \tag{35}$$

Which means

$$U^\dagger XU = \left(x_i + \theta^{-1}\sin(\theta)(\varepsilon_{jki}\theta_j x_k) - \theta^{-2}(1-\cos(\theta))\theta_j\theta_i x_j\right)\sigma_i \tag{36}$$

Which is exactly the Rodrigues Rotation formula, meanign the vector x_i transforms like a vector in 3d under an $SO(3)$ transformation.
This is the $SU(2) \to SO(3)$ mapping

$$U \in SU(2) \sim O \in SO(3)\ if\ U^\dagger XU = (O\vec{x})\cdot\vec{\sigma} \tag{37}$$

Since U appears twice in the equation, the same matrix O maps to both U and $-U$, making this mapping a surjective two-to-one homomorphism.

2 A bit of $SO(1,3)$

2.1 Weyl Spinor and Lorentz transformation relations

Using the hist, let us first show that any Pauli matrix σ_n satisfies $\sigma_n\sigma_2 = -\sigma_2\sigma_n^T$. The easiest way to show this is just to calculte both sides for each of the matrices and see that they give the same result. We will use the identities $\sigma_i\sigma_j = i\varepsilon_{ijk}\sigma_k + \delta_{ij}\mathbb{1}$ and $\sigma_1^T = \sigma_1, \sigma_2^T = -\sigma_2$, and $\sigma_3^T = \sigma_3$.

$$\begin{aligned} \sigma_1\sigma_2 = i\sigma_3 \qquad & -\sigma_2\sigma_1^T = -\sigma_2\sigma_1 = -(-i\sigma_3) = i\sigma_3 \\ \sigma_2\sigma_2 = \mathbb{1} \qquad & -\sigma_2\sigma_2^T = \sigma_2\sigma_2 = \mathbb{1} \\ \sigma_3\sigma_2 = -i\sigma_1 \qquad & -\sigma_2\sigma_3^T = -\sigma_2\sigma_3 = -(i\sigma_1) = -i\sigma_1 \end{aligned} \tag{38}$$

This property is true, not only for σ_n, but also for σ_n^T, since they are just σ_n up to a constant.
Using the representation we found in 13:

$$\begin{aligned} M_L^T\sigma_2 = \exp\Big(-i\left(\theta_n - iy_n\right)\frac{\sigma_n}{2}\Big)^T\sigma_2 = \left[\cos\left(\frac{\sqrt{\theta^2+y^2}}{2}\right)\mathbb{1} - \sin\left(\frac{\sqrt{\theta^2+y^2}}{2}\right)\frac{\theta_n - iy_n}{\sqrt{\theta^2+y^2}}\sigma_n^T\right]\sigma_2 = \\ = \left[\cos\left(\frac{\sqrt{\theta^2+y^2}}{2}\right)\sigma_2 + \sin\left(\frac{\sqrt{\theta^2+y^2}}{2}\right)\frac{\theta_n - iy_n}{\sqrt{\theta^2+y^2}}\sigma_2\sigma_n\right] = \\ = \sigma_2\left[\cos\left(\frac{\sqrt{\theta^2+y^2}}{2}\right)\mathbb{1} + \sin\left(\frac{\sqrt{\theta^2+y^2}}{2}\right)\frac{\theta_n - iy_n}{\sqrt{\theta^2+y^2}}\sigma_n\right] = \\ = \sigma_2\exp\Big(i\left(\theta_n - iy_n\right)\frac{\sigma_n}{2}\Big) = \sigma_2 M_R^\dagger \end{aligned} \tag{39}$$

It is also easy to see that

$$M_R^\dagger M_L = \exp\Big(i\left(\theta_n - iy_n\right)\frac{\sigma_n}{2}\Big)\exp\Big(-i\left(\theta_n - iy_n\right)\frac{\sigma_n}{2}\Big) = \mathbb{1} \tag{40}$$

Which is to say $M_R^\dagger = M_L^{-1}$ and so

$$M_L^T\sigma_2 = \sigma_2 M_R^\dagger = \sigma_2 M_L^{-1} \tag{41}$$

2.2 Lorentz scalar from the spinor fields

The Lorentz transformaiton for the vector A^μ is $A^\mu \to A'^\mu = \Lambda^\mu_\nu A^\nu$. We saw in section 1 that this is equivilant to transform the matrix $A^\mu\bar{\sigma}_\mu \to M_R A^\mu\bar{\sigma}_\mu M_R^\dagger$. We saw in class that $\bar{\sigma}_\mu$ transforms using $U = M_R^\dagger$. Using this, together with the transformations given for the spinor fields in the question, we get

$$\xi^\dagger A^\mu\bar{\sigma}_\mu\xi \to \xi^\dagger M_L^\dagger M_R A^\mu\bar{\sigma}_\mu M_R^\dagger M_L\xi = \xi^\dagger A^\mu\bar{\sigma}_\mu\xi \tag{42}$$

There is, of course, a similar scalar for the χ field, simply by replacing $\xi \to \chi, \bar{\sigma}_\mu \to \sigma_\mu$. Then, $\sigma_\mu \to M_L\sigma_m u M_L^\dagger$:

$$\chi^\dagger A^\mu\sigma_\mu\chi \to \chi^\dagger M_R^\dagger M_L A^\mu\sigma_\mu M_L^\dagger M_R\chi = \chi^\dagger A^\mu\sigma_\mu\chi \tag{43}$$

In both cases we used $M_L^{-1} = M_R^\dagger$.

3 What adds under Lorentz transformations?

3.1 Rotations

For a rotation around the z-axis, the 4×4 Lorentz transformation is

$$\Lambda^{\mu}_{\nu}(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{44}$$

Using the well known trigonometric identities for angle addition, it's easy to see that performing two successive rotations around the z-axis by two angles θ_1 and θ_2 is equivilant to a single rotation by $\theta_1 + \theta_2$

$$\begin{gathered} \Lambda^{\mu}_{\alpha}(\theta_1)\Lambda^{\alpha}_{\nu}(\theta_2) = \\ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta_1 & -\sin\theta_1 & 0 \\ 0 & \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta_2 & -\sin\theta_2 & 0 \\ 0 & \sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \\ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 & -\cos\theta_1\sin\theta_2 - \sin\theta_1\cos\theta_2 & 0 \\ 0 & \sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2 & \sin\theta_1\sin\theta_2 + \cos\theta_1\cos\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \\ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) & 0 \\ 0 & \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \Lambda^{\mu}_{\nu}(\theta_1+\theta_2) \end{gathered} \tag{45}$$

3.2 Boosts

A boost along the x-axis by a velocity parameter β is

$$\Lambda^{\mu}_{\nu}(\beta) = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{46}$$

Where $\gamma = \left(1-\beta^2\right)^{-1/2}$. Performing two consecutive such transformations by two different velocities β_1 and β_2, will amount to

$$\begin{gathered} \Lambda^{\mu}_{\alpha}(\beta_1)\Lambda^{\alpha}_{\nu}(\beta_2) = \\ = \begin{pmatrix} \gamma_1 & -\beta_1\gamma_1 & 0 & 0 \\ -\beta_1\gamma_1 & \gamma_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma_2 & -\beta_2\gamma_2 & 0 & 0 \\ -\beta_2\gamma_2 & \gamma_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \\ = \begin{pmatrix} \gamma_1\gamma_2(1+\beta_1\beta_2) & -\gamma_1\gamma_2(\beta_1+\beta_2) & 0 & 0 \\ -\gamma_1\gamma_2(\beta_1+\beta_2) & \gamma_1\gamma_2(1+\beta_1\beta_2) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{gathered} \tag{47}$$

Let us now look for what β and γ fit this new transformation. This means we are looking for β and its matching γ for which $\Lambda^{\mu}_{\alpha}(\beta_1)\Lambda^{\alpha}_{\nu}(\beta_2) = \Lambda^{\mu}_{\nu}(\beta)$, so we can find the addative connection. From the last matrix, we can see that

$$\gamma = \gamma_1\gamma_2(1+\beta_1\beta_2) \qquad \beta\gamma = -\gamma_1\gamma_2(\beta_1+\beta_2) \tag{48}$$

Therefore

$$\beta = \frac{\gamma\beta}{\gamma} = \frac{\gamma_1\gamma_2(\beta_1+\beta_2)}{\gamma_1\gamma_2(1+\beta_1\beta_2)} = \frac{\beta_1+\beta_2}{1+\beta_1\beta_2} \tag{49}$$

This reminds us of the addition formula for hyperbolic tangent:

$$\tanh(\alpha+\beta) = \frac{\tanh\alpha + \tanh\beta}{1+\tanh\alpha\tanh\beta} \tag{50}$$

So let us define the rapidity, with a range $(-\infty, \infty)$:

$$y(\beta) = \tanh^{-1}(\beta) \tag{51}$$

and thus equation 49 becomes

$$\tanh(y) = \frac{\tanh(y_1)+\tanh(y_2)}{1+\tanh(y_1)\tanh(y_2)} \tag{52}$$

$$y = \tanh^{-1}\left(\frac{\tanh(y_1)+\tanh(y_2)}{1+\tanh(y_1)\tanh(y_2)}\right) = \tanh^{-1}(\tanh(y_1+y_2)) = y_1+y_2 \tag{53}$$

Which means

$$\Lambda^\mu_\alpha(y_1)\Lambda^\alpha_\nu(y_2) = \Lambda^\mu_\nu(y_1+y_1) \tag{54}$$

4 Translation in field space

4.1 Translation using the Noether charge

Opening the equation using Campbell identity

$$e^{i\alpha Q}\phi(x)e^{-i\alpha Q} = \sum_{n=0}^{\infty}\frac{1}{n!}\left[i\alpha Q, \phi(x)\right]_n \tag{55}$$

Where

$$[A,B]_n = [A,[A,B]_{n-1}] \qquad [A,B]_0 = B \tag{56}$$

Specifically, for the charge $Q = \int d^3x\pi$, we can use the commutation relations

$$[\phi(x), \pi(y)] = i\delta^3(x-y) \tag{57}$$

Which means

$$[i\alpha Q, \phi(x)] = i\alpha\int d^3y[\pi(y),\phi(x)] = i\alpha\int d^3y(-i)\delta^3(x-y) = \alpha \tag{58}$$

And so every commutation relations after the second order vanish, and we are left with

$$e^{i\alpha Q}\phi(x)e^{-i\alpha Q} = \phi(x)+\alpha \tag{59}$$

4.2 Additional Lorentz invariants

Any other Lorentz invariant that is only dependent on the derivative of the field is also invariant to an addition of a constant to the field. The most simple function of derivatives is polynomials, and the next polynomial that is not already in the Lagrangian is the quadruple derivative:

$$S = \int d^4x\frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{1}{4}\left(\partial_\mu\phi\partial^\mu\phi\right)^2 \tag{60}$$

4.3 The new Conjugate momentum

The new conjugate momentum is

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = [1 + (\partial_\mu \phi \partial^\mu \phi)]\dot{\varphi} \tag{61}$$

Using this new momentum, we can see if the shift in the field space is stil the same. For this we will calculate the commutation relations:

$$[\phi(x), \pi(y)] = [\dot{\varphi} + \partial_\mu^{(x)} \phi \partial^{(x)\mu} \phi \dot{\varphi}, pi(y)] = i\delta^3(x-y)(1 + \partial_\mu^{(x)} \partial^{(x)\mu}) \tag{62}$$

And thus, since the derivative of 1 vanishes, we get:

$$[i\alpha Q, \phi(x)] = i\alpha \int d^3y [\pi(y), \phi(x)] = i\alpha \int d^3y(-i)\delta^3(x-y)(1 + \partial_\mu^{(x)} \partial^{(x)\mu}) = \alpha \tag{63}$$

Meaning that once again, the Campbell identity gets cut off after the linear term in α, and the momentum once again shifts the field by a constant.

5 The Baker-Campbell-Hausdorff formula

5.1 Proving for the first and second order

Opening the exponents up to second order:

$$\begin{aligned} e^A e^B &= \left(1 + A + \frac{1}{2}A^2 + O(A^3)\right)\left(1 + B + \frac{1}{2}B^2 + O(B^3)\right) = \\ &= 1 + A + B + \frac{1}{2}A^2 + AB + \frac{1}{2}B^2 + \cdots = \\ &= 1 + (A+B) + \frac{1}{2}(A+B)^2 + \frac{1}{2}[A,B] + \cdots = \\ &= \exp\left(A + B + \frac{1}{2}[A,B] + \cdots\right) = e^C \end{aligned} \tag{64}$$

As for the third order, there each term requires a different commutator, since the third term in the exponent of $A+B$ is

$$\frac{1}{6}(A+B)^3 = \frac{1}{6}\left(A^3 + A^2B + AB^2 + B^3 + B^2A + BA^2 + ABA + BAB\right) \tag{65}$$

While the third order of the result from the product of the exponent $e^A e^B$ is

$$\frac{1}{2}AB^2 + \frac{1}{2}A^2B + \frac{1}{6}A^3 + \frac{1}{6}B^3 \tag{66}$$

Subtracting them will give us the remainder, which is the third order term in C:

$$\begin{aligned} &\frac{1}{2}AB^2 + \frac{1}{2}A^2B + \frac{1}{6}A^3 + \frac{1}{6}B^3 - \frac{1}{6}\left(A^3 + A^2B + AB^2 + B^3 + B^2A + BA^2 + ABA + BAB\right) = \\ &= \frac{1}{6}\left(2AB^2 - A^2B - ABA + 2BA^2 - B^2A - BAB\right) = \\ &= \frac{1}{6}\left([AB, B] + [BA, A] + [A, B^2] + [B, A^2]\right) \end{aligned} \tag{67}$$

Which means that the third order of the approximation is also made of commutators.

5.2 The $SL(2,C)$ group and the divergence of the identity

The gropu $SL(2,C)$ is defined as the group of all complex $2x2$ matrices with determinant 1. Now, let us take a matrix $S \in SL(2,C)$ very close to the identity:

$$S \approx \mathbb{1} + \epsilon = \begin{pmatrix} 1+\epsilon_{11} & \epsilon_{12} \\ \epsilon_{21} & 1+\epsilon_{22} \end{pmatrix} \tag{68}$$

This matrix must satisfy $\det S = 1$, and so:

$$(1+\epsilon_{11})(1+\epsilon_{22}) - \epsilon_{12}\epsilon_{21} = 1 + \epsilon_{11} + \epsilon_{22} + \epsilon_{11}\epsilon_{22} - \epsilon_{12}\epsilon_{21} = 1 + \mathrm{tr}(\epsilon) + \det(\epsilon) = 1 \tag{69}$$

Since $\det \epsilon$ is of the second order, we can neglect it and get

$$\mathrm{tr}\,\epsilon = 0 \tag{70}$$

Or, in other words

$$\epsilon_{11} = -\epsilon_{22} \tag{71}$$

Which means that a simple basis of three traceless matrices for the algebra is

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \tag{72}$$

Now we can look at the matrices X and Y from $SL(2,C)$:

$$X = \begin{pmatrix} 0 & i\pi \\ i\pi & 0 \end{pmatrix} \qquad Y = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \tag{73}$$

To calcualte their exponents, we will first calcualte their powers:

$$X^2 = -\pi^2 \mathbb{1} \qquad Y^2 = 0 \tag{74}$$

Therefore:

$$\begin{aligned} e^X &= \sum_n \frac{X^n}{n!} = \sum_{n\ even} \frac{X^n}{n!} + \sum_{n\ odd} \frac{X^n}{n!} = \sum_{n\ even} \frac{(i\pi)^n}{n!}\mathbb{1} + \sum_{n\ odd} \frac{(i\pi)^{n-1}}{n!} X = \\ &= \cos(\pi)\mathbb{1} + \frac{1}{\pi}\sin(\pi) X = -\mathbb{1} \end{aligned} \tag{75}$$

Which is, of course, part of the group, and

$$e^Y = \sum_n \frac{X^n}{n!} = \mathbb{1} + Y = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \tag{76}$$

which is also a part of the group. However:

$$e^X e^Y = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \tag{77}$$

The determinant of which is $\det\left(e^X e^Y\right) = 1$, which means it is a part of $SL(2,C)$. However, if we try to find the matrix Z such that $e^Z = e^X e^Y$, we will find that we can't. The proof goes as follows:
The matrix Z has two eigenvalues which could be differen or equal. If they are different, they must satisfy $\lambda_1 = -\lambda_2$, since $\mathrm{tr}\,Z = 0$. This would mean Z is diagonalizable and thus e^Z is also diagonalizable. If the two eigenvalues are equal, then they must be zero, which means that there exist a vector $\vec{v}$ such that $e^Z\vec{v} = e^0\vec{v} = 1\vec{v} = \vec{v}$, meaning that 1 must be an eigenvalue of the matrix e^Z. We got a conclusion that every matrix that can be displayed as e^Z is either diagonalizable or has an eigenvalue 1. Since the matrix that we got is doesn't satisfy any of these requirements, it canno't be represented in this form.